



DEBUG DIARIES

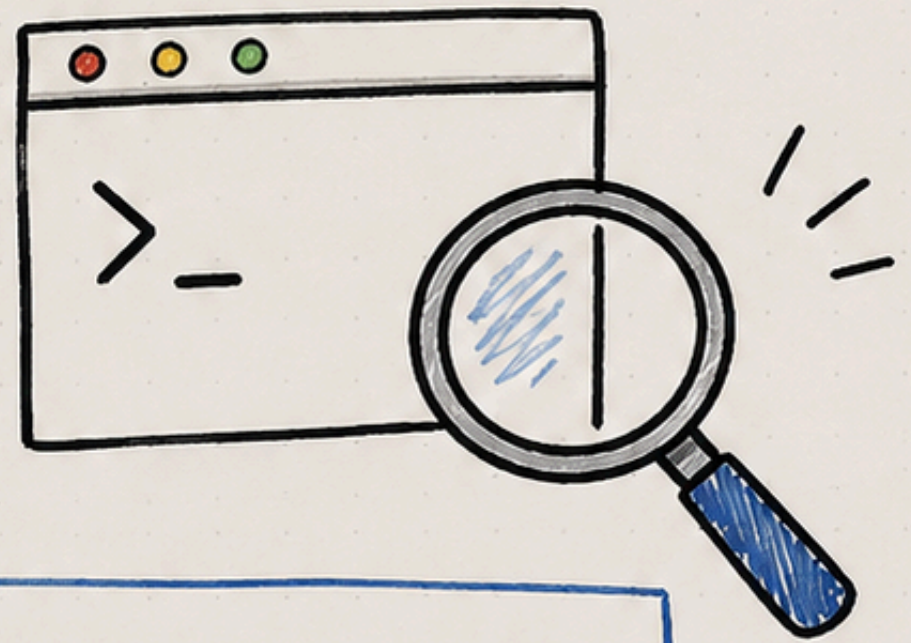
#1

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5 Linux Troubleshooting Scenarios

Every Beginner Should Know

Can you solve them
before looking at
the answer?



These are real situations you might
face in interviews or while working
on Linux servers.

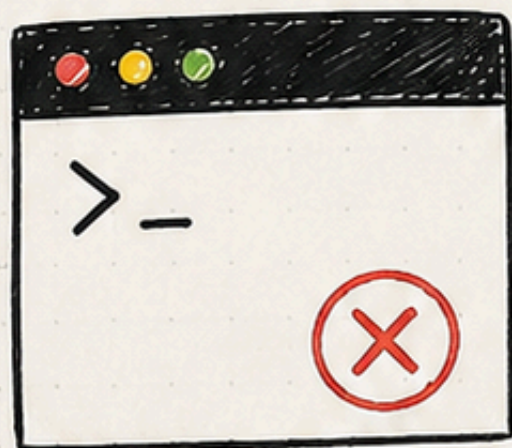
→ Try solving each one before swiping
to the solution.





SCENARIO 1

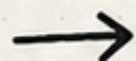
Service Not Starting



You rebooted the server.
Your web app service
'myapp' is not running.
What do you do first?

Think like this:

Is the service
running or
failed?



What do the
logs say?



Is it enabled
to start on
boot?

Your turn



List 4 commands (in order)
you would run to diagnose
this issue.

①

②

③

④

HINTS



- Start with checking the service status.
- Then check the logs.
- Finally, check if it's enabled on boot.

Commands to explore:

- `systemctl status myapp`
- `journalctl -u myapp -n 50`
- `systemctl is-enabled myapp`



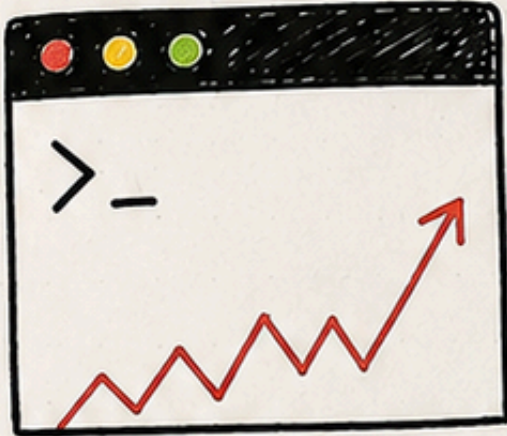
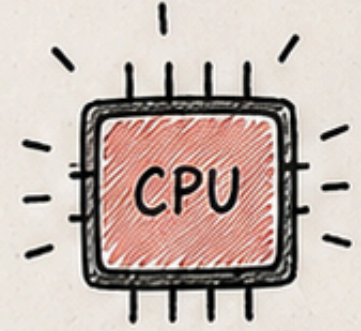
Don't restart blindly. Understand first.





SCENARIO 2

High CPU Usage



Your manager says,
"The server is very slow."
You SSH into the server.
What will you do first to
find the cause?

Goal:

Find which
process is
using high
CPU.

Your turn!

? List 3-4 commands (in order)
you would run to identify
which process is using
high CPU.

- ① _____
- ② _____
- ③ _____
- ④ _____

HINTS

- ★ Use a command that shows live CPU usage.
- ★ Look for processes sorted by CPU %.
- ★ Note the PID (Process ID) of the top process.

Commands to explore:

- `top` (press 'q' to quit)
- `htop` (more user-friendly)
- `ps aux --sort=-%cpu | head -10`
(shows top 10 by CPU %)



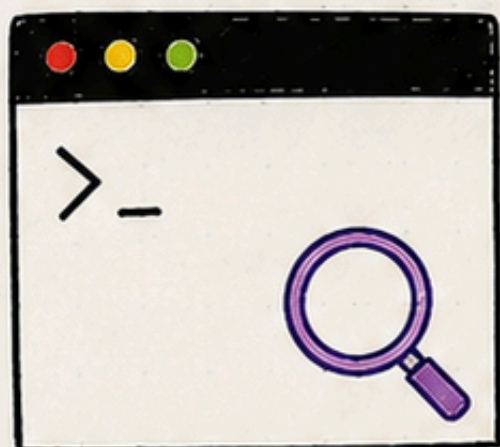
Find the process first.
Then decide what to do.

Don't kill blindly! 😊



SCENARIO 3

Finding Service Logs

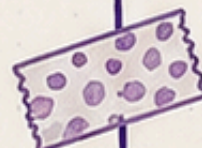


A developer asks:

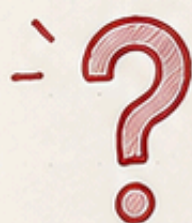
"Where are the logs for the 'docker' service?"

The service is managed by systemd.

What commands would you use?



Your turn!



List 3 commands you would use to check the logs for a systemd service.

- ① _____
- ② _____
- ③ _____

HINTS

- ★ systemd services → logs are stored in journald.
- ★ Use -n to limit number of lines.
- ★ Use -f to follow logs in real-time (like tail -f).

COMMANDS TO EXPLORE

- ① `systemctl status <service>`
Check service status first
- ② `journalctl -u <service> -n 50`
View last 50 lines of logs
- ③ `journalctl -u <service> -f`
Follow logs in real-time



Logs are your best friend.

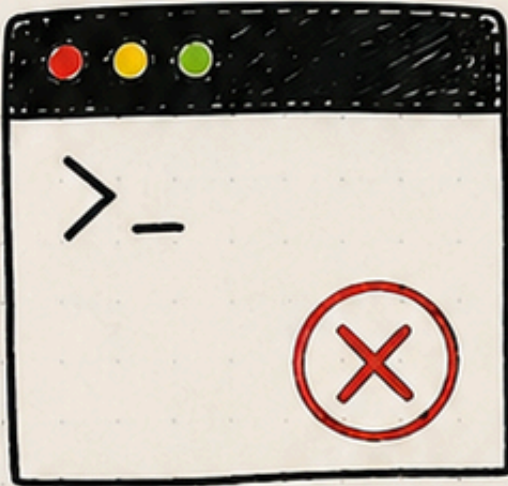
They usually tell you exactly what went wrong.





SCENARIO 4

File Permission Denied



You try to run a script:

```
./backup.sh
```

But Linux says:

Permission denied

What would you do first?

Your turn!



List 3-4 steps (in order)
you would take to
fix this.

① _____

② _____

③ _____

④ _____

HINTS

- ★ Check the file permissions.
- ★ Does the file have execute (x) permission?
- ★ Add execute permission if needed.
- ★ Verify and try again.



COMMANDS TO EXPLORE

- ①

```
ls -l backup.sh
```


Check current permissions
- ②

```
chmod +x backup.sh
```


Add execute permission
- ③

```
ls -l backup.sh
```


Verify changes
- ④

```
./backup.sh
```


Try running again



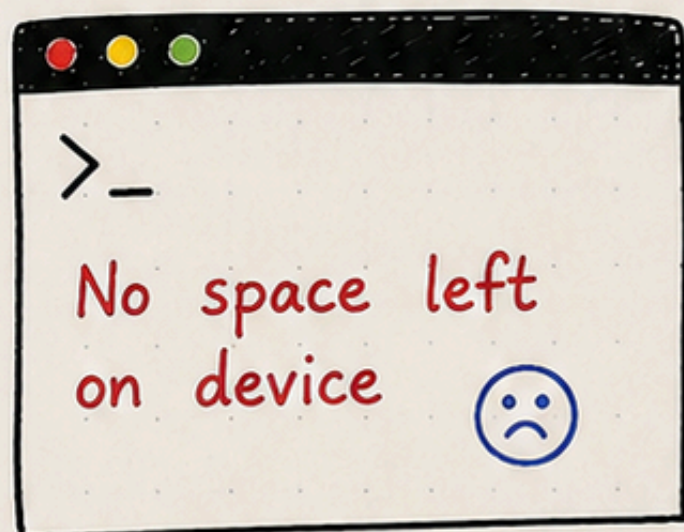
Remember: Files need 'x' (execute) permission to run. Don't forget to check!





SCENARIO 5

Disk Space Full !



Your application crashes.

You check the logs and see:

No space left on device

What commands will you run to find what's eating up space?

Your turn!



List 3-4 commands (in order) you would use to find what's using up disk space.

- ① _____
- ② _____
- ③ _____
- ④ _____

HINTS

- ★ Find overall disk usage of all filesystems.
- ★ Find large directories.
- ★ Drill down to find large files.
- ★ Sort and check the biggest ones first.



COMMANDS TO EXPLORE

- ① `df -h` # Check disk usage of filesystems
- ② `du -sh /*` # Find size of directories in /
- ③ `du -h /path | sort -hr | head -10`
Find biggest directories in a path
- ④ `du -h /path/to/dir | sort -hr | head -10`
Drill down to biggest files



Full disk = serious problems.
Smart engineers find the cause, not just the error.





WRAP-UP

Troubleshoot Like a Pro (Not Like a Panicked Penguin)



No matter the issue, follow this mindset.



- ① Understand the problem
- ② Form a hypothesis
- ③ Gather information (logs, status, usage, etc.)
- ④ Narrow down the cause
- ⑤ Fix it. Verify it. Learn from it.

★ Linux won't tell you what's wrong. You have to ask the right questions.

KEY TAKEAWAYS

Don't guess.
Investigate!



Logs are
your best
friend.



Check status
before digging
deep.



Small commands
solve big
problems.



Stay calm.
Think.
Then fix.



KEEP PRACTICING

The more you troubleshoot,
the better you get.
Today it's practice,
tomorrow it's confidence!



That's it for Debug Diaries #1!
Hope this was helpful.
See you in the next one
with more real-world scenarios.



You got
this! ❤️



Learn. Practice. Debug. Repeat.

